

Annabel Wade

Boston, MA | **Email:** wade@bu.edu | **GitHub:** annabelwade

- Diligent PhD Student at Boston University's Faculty of Computing & Data Sciences studying the training dynamics of AI weather models.
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Education

PhD in Environmental Data Science | *Boston University, MA* | Sept 2025 – May 2030 (Expected)

B.S. in Applied Mathematics: Data Science | *University of Washington, WA* | Sept 2021 – March 2025

- **Minor:** Oceanography
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Research Experience

Research Assistant | *University of Washington School of Oceanography, Seattle, WA* | March 2025 – Aug 2025

- Developed and analyzed global ocean geostrophic velocity product with statistical mapping and uncertainty quantification, comparing to other datasets and creating insightful visualizations.
- Co-authored a manuscript that is in preparation.

Hollings Intern | *NOAA Global Systems Laboratory, Boulder, CO* | June 2024 – March 2025

- Training deep learning models for detection of wildfire smoke from satellite imagery with Python-based high performance computing.
- Produced an ensemble of models that improved boundaries of detected smoke plumes and generalization.
- Published a first-author paper at ICLR 2026 Machine Learning for Remote Sensing workshop.

Research Assistant | *University of Washington School of Oceanography, Seattle, WA* | June 2022 – June 2024

- Developed an interactive data visualization tool in Python that accelerated team progress and improved communication of scientific insights.
 - Performed multiscale spatial-temporal mapping and quality control of ocean data products.
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Teaching Experience

Research Mentor | *Boston University* | March 2026 – May 2026

- Mentored 3 data science undergraduate students on a project perturbing initial conditions to reveal when an AI weather model learns physical relationships.
- Taught students how to run parallel model inference using earth2studio and PyTorch on a high-performance computing cluster, perform rigorous scientific analysis, and communicate results in final poster and talk presentations.

Instructional Assistant | *University of Washington Youth and Teens Programs* | Aug 2024 – Sept 2024

- Supported two classes of 20-30 high school students each through data-driven modeling projects.
- Taught six lectures on high-school level mathematics for regression modeling.

Teaching Assistant | *University of Washington School of Oceanography* | Sept 2022 – March 2023

- Set up and explained weekly labs for undergraduate Fluid Mechanics and graduate Geophysical Fluid Dynamics.
 - Guided a class of 30 students through the application of physics concepts via experiential learning.
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Publications

Accepted Papers

- **A. Wade, R. Koki, C. Kumler-Bonfanti, D. Durran. "Improving Satellite-based Wildfire Smoke Plume Detection with Deep Ensembles."** *International Conference on Learning Representations (ICLR) 2026 Machine Learning for Remote Sensing workshop.*
- **R. Koki, M. McCabe, D. Kedar, J. Myers-Dean, A. Wade, et al. "SmokeViz: A large-scale satellite dataset for wildfire smoke detection and segmentation."** *Conference on Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track 2025.*

In-Preparation

- **A. Wade, N. Saphra, E. A. Barnes. "Beyond the final checkpoint: training dynamics reveals insight into how a weather emulator learns physics"**
- **A. Gray, C. Wortham, O. Kihle, K. Christensen, A. Wade. "A framework for multiscale interpolation and data fusion: an application to global observation-based ocean velocities."**
- **M. Chien*, V. Munteanu*, A. Wade*, E. A. Barnes. "Physical consistency in AI climate emulators."**

(* = equal contribution)

Selected Presentations

- **Beyond the final checkpoint: studying the training dynamics of an AI weather model.** Poster presentation presented at the IMSI Workshop “Unpacking Weather Emulators” (September 2026).
 - **Training dynamics of a weather emulator: an introductory atmospheric river case study.** Lighting talk presented at the BU Computing & Data Sciences PhD Student Lighting talks (April 2026).
 - **Exploring Multi-Model Ensembling Methods to Improve on Deep Learning Based Smoke Detection.** Talk presented at the 105th AMS Annual Meeting, New Orleans, LA (Jan 2025).
Awarded 3rd place for Best Student Oral Presentation.
 - **Estimating Global Ocean Absolute Geostrophic Velocities with Satellite and Autonomous Float Observations.** Poster presented at Ocean Sciences Meeting 24, New Orleans, LA (Feb 2024).
 - **Estimating Global Ocean Eddy Kinetic Energy with Autonomous Floats and Satellites.** Talk presented at Washington NASA Space Grant Undergraduate Research Symposium (Aug 2023).
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Awards

- **NASA Space Grant Scholarship** | 2024 – 2025
 - **NOAA Ernest F. Hollings Scholarship** | 2023 – 2025
 - **Academic Excellence Award**, UW Applied Mathematics | 2024
 - **Undergraduate Research Funding**, Washington NASA Space Grant | 2023
 - **Rita A. Colwell Scholarship for Women in Oceanography** | 2022 – 2023
 - **Lowell K. and Alice M. Barger Scholarship** | 2021 – 2023
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Outreach

Graduate Climate Conference | Organizer | 2026 - present

- Co-chaired oral session on Climate and Atmospheric Modeling.
- Co-chaired the Audio/Visual committee, which manages all presentation and audio tech during the conference.
- Supported advertising and abstract evaluation during conference preparation.

UW Program on Climate Change | Co-Leader | 2021 – 2023

- Led seminars and panels to involve undergraduates with climate-related research.

Bryant Elementary School | Science Fair Mentor | 2023

- Guided 5th-grade students through the scientific method and environmental data analysis